Fundamentals of Deep Learning and Programming

Lecture 0: Course Overview

Fall 2025

Adjunct Professor: Donghoon Kang

Course Overview

Lecture Time	Fridays, 2 PM – 5 PM
Classroom	Room: L3 301, 302 (3334A, B)
Evaluation Criteria	Attendance (10%) Programming Assignments (15%) Term-project (15%) Mid-term (30%), Final (30%)

- Programming assignments: students should submit the code and a 2-3 page document briefly explaining the code to me by email (chocopie@kist.re.kr)

Supplementary Class

오늘 < 2025.10 > 음력 손없는날 기념일

일	월	화	수	목	금	토
28	29	30	1 국군의 날	2 노인의 날	3 개천절	4 이산가족...
5 세계 한인...	6 음 8.15 추석	7	8 대체 휴일 한로 재향 군인...	9 한글날	10 임산부의 날 정신건강...	11 호스피스...
12	13	14	15 체육의 날	16 세계 식량... 부마민주...	17	18 문화의 날
19	20	21 음 9.1 경찰의 날	22	23 상강	24 국제연합일	25 독도의날
26	27	28 금융의 날 교정의 날	29 지방자치...	30	31 회계의 날	1

Holiday 10.03 (Friday)

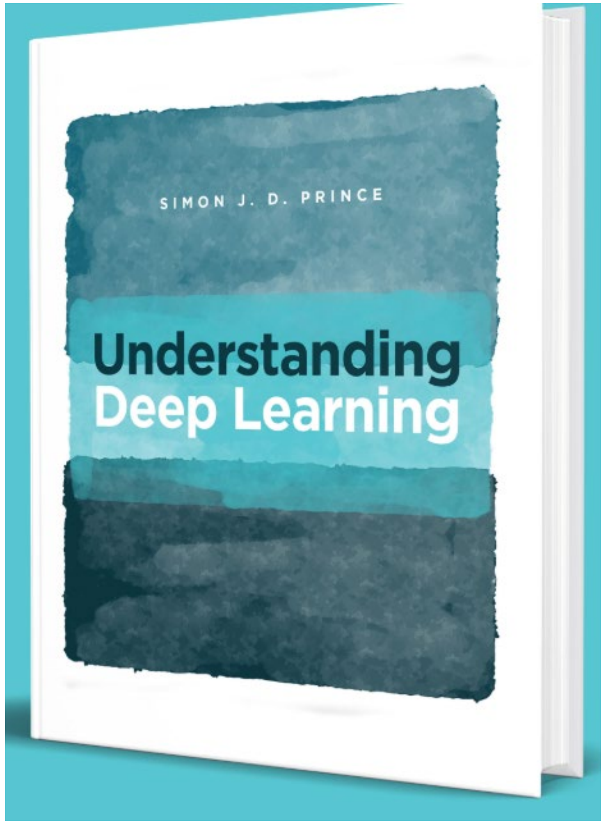
Thursday 4 PM ~ 6:30 PM (?)

Course Website

`https://kimbabmoowoo.github.io/lecture.html`

Recommended book (pdf is free)

`https://udlbook.github.io/udlbook/`



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by Simon J.D. Prince | Dec 5, 2023
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5 star	94%
4 star	3%
3 star	1%
2 star	0%
1 star	2%

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Term Project

Must be performed alone

- Should select a paper published only in CVPR, ICCV, ECCV, or NeurIPS after 2022 (inclusive).
- Should write an 8-10 page report analyzing the "training code" of the selected paper.

Before Week 7	Should decide which paper to choose. (also possible to change another paper later)
Before Week 15	Should submit an 8-10 page report as a term-project.

Schedule

September

Week 1	Course overview Introduction to Python Programming
Week 2	Useful Python Libraries — Numpy, PyTorch
Week 3	Neural Networks Basics
Week 4	Convolutional Neural Networks (CNNs)

Schedule

October

Week 5	Models using CNN
Week 6	Introduction to Natural Language Processing (NLP)
Week 7	Attention Mechanism
Week 8	Transformer — Part 1
Week 9	Mid term exam

Schedule

November

Week 10	Transformer	Part 2	
Week 11	Transformer	Part 3	Vision Transformer (ViT)
Week 12	Generative Models – Introduction to GANs, VAEs		
Week 13	Diffusion Models	Part 1	

Schedule

December

Week 14	Diffusion Models	Part 2
Week 15	Diffusion Models	Part 3
Week 16	Final exam	